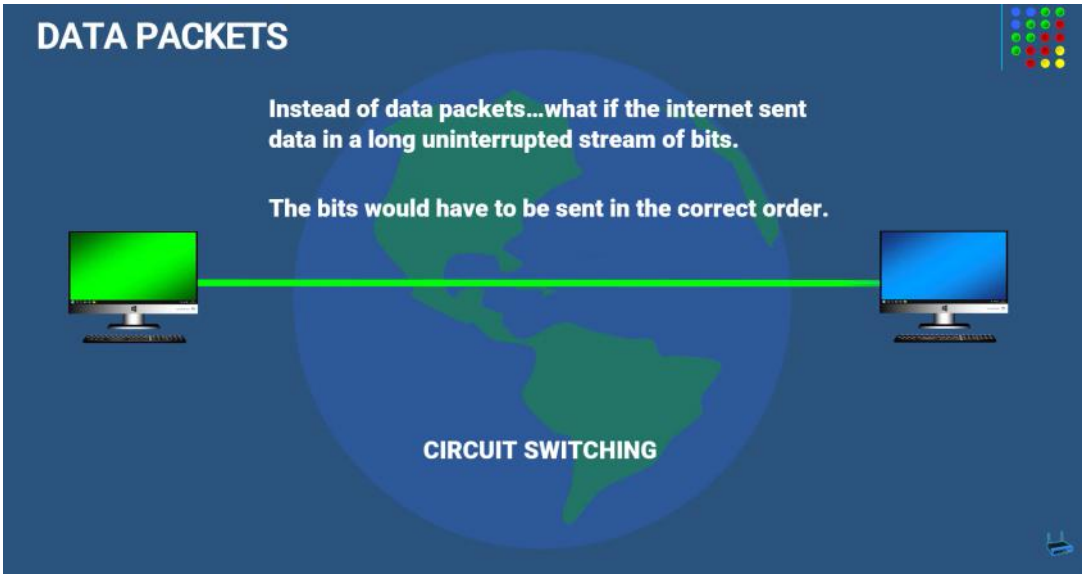
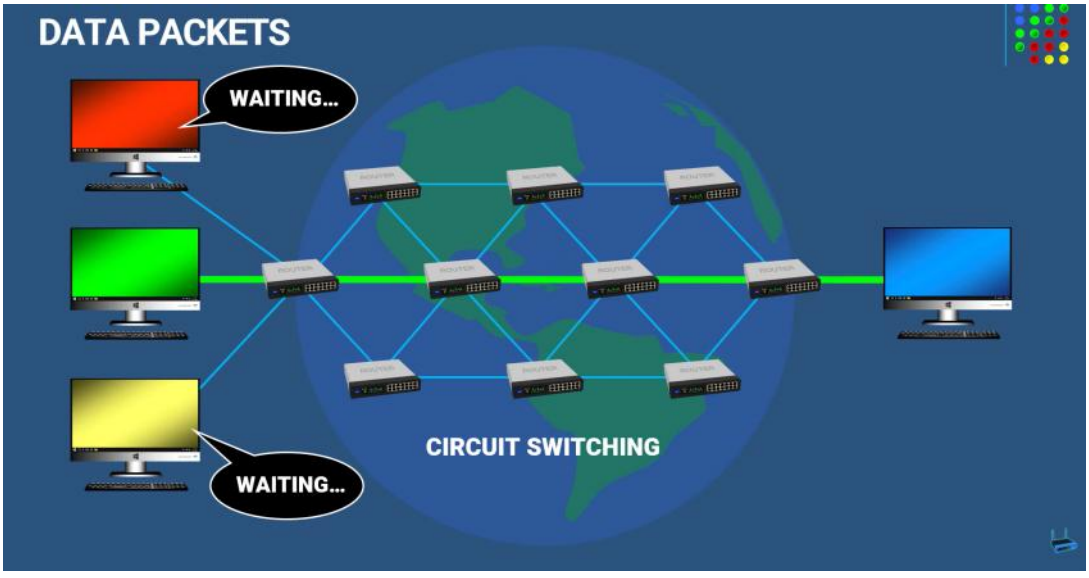
Data Packets

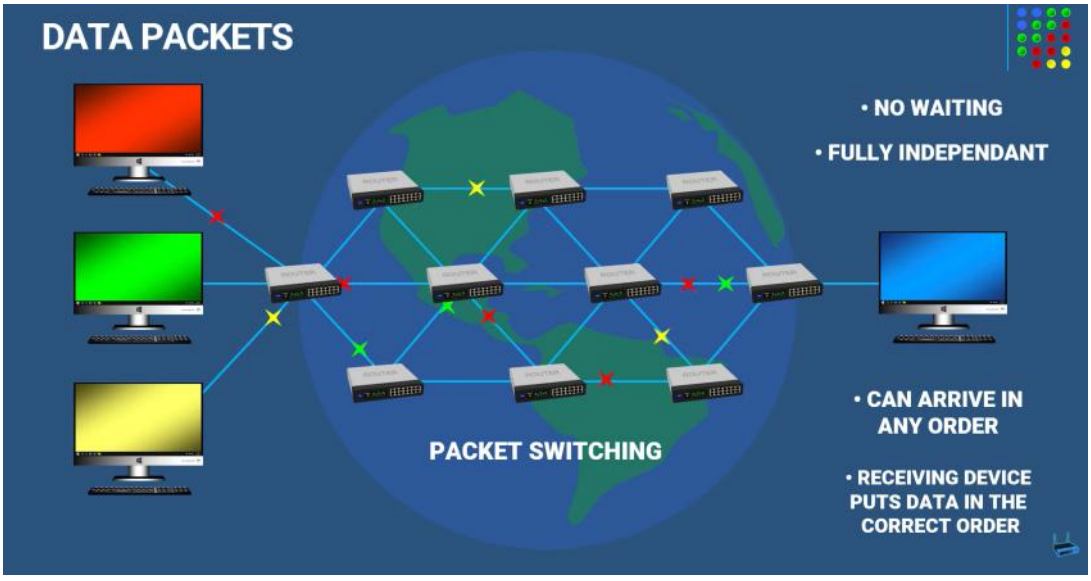
Wednesday, April 22, 2026 7:19 AM

Data on the internet is not transmitted in one single piece. Instead, it uses the TCP/IP protocol to break files—such as emails, photos, or videos—into smaller, independent segments known as data packets.

Packet switching is more efficient as the device does not have to wait for some time to complete the whole transfer of data at a time if it was using circuit switching.

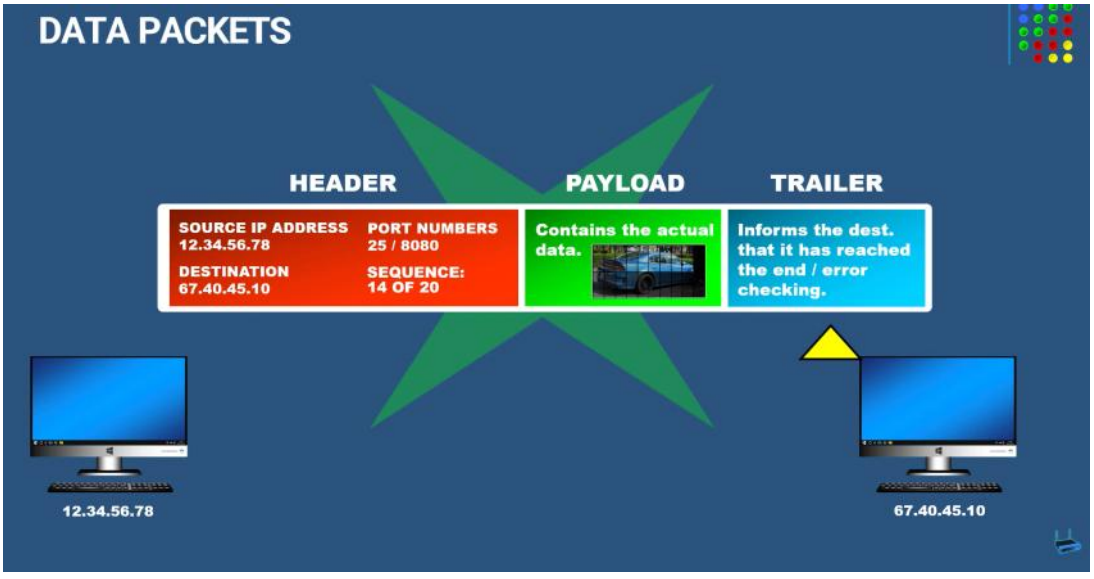


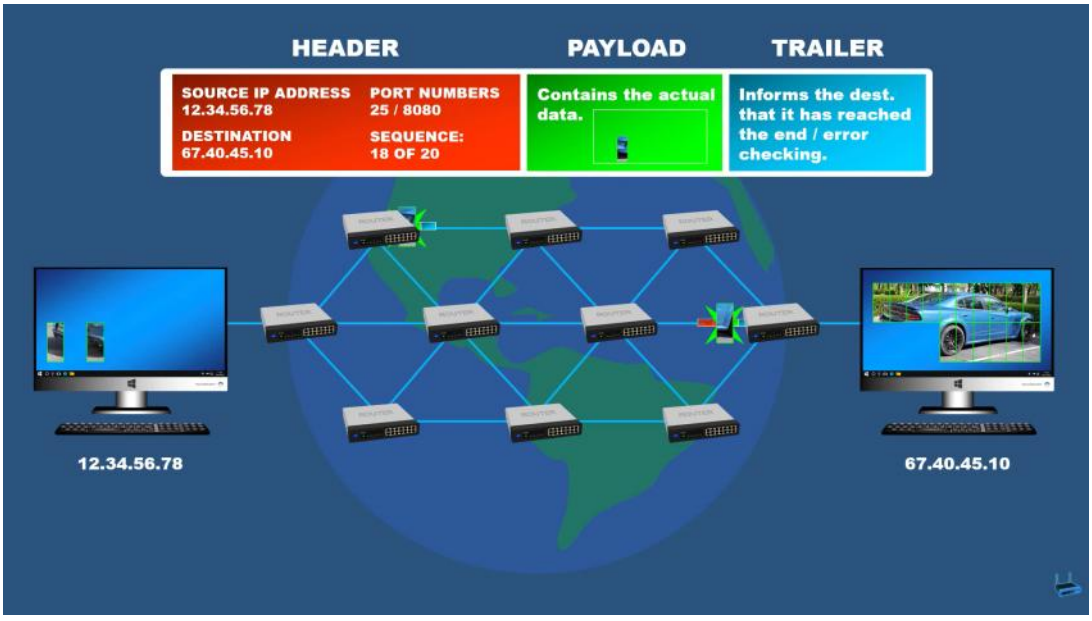




This method, called packet switching, is more efficient than circuit switching because it prevents data bottlenecks. Each packet contains essential information, including:

- Header: Holds the source and destination IP addresses, port numbers, and sequence numbers to ensure correct reassembly.
- Payload: Contains the actual chunk of data being sent.
- Trailer: Signals the end of the transmission and checks for errors.
- Because each packet is independent, they can travel across various network routes simultaneously. Once they arrive at their destination, the receiving device uses the sequence numbers to reassemble them into the original file and verifies that the data is error-free.




HEADER
PAYLOAD
TRAILER
SOURCE IP ADDRESS
12.34.56.78
PORT NUMBERS
25 / 8080
DESTINATION
67.40.45.10
SEQUENCE:
18 OF 20
Contains the actual data.
Informs the dest. that it has reached the end / error checking.
12.34.56.78
67.40.45.10